

Conformal Blocks and Crossing Symmetry

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Abstract

Here is a note on a AMAZING topic of 2 dimensional CFT developed in the 80s.

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1 Introduction

1.1 Overview

Since the BPZ paper gives a concrete definition of the conformal blocks, we want to control the behavior of these special functions. A series of papers by Moore and Seiberg in the late 80s and early 90s give an answer. Without concretely calculating the conformal blocks, they show that these function have certain crossing properties, controlled by series of duality matrices. More amazingly, these duality matrices can be explicitly calculated. This gives a explicit way of bootstrapping the 2D CFT structure constants.

Moreover, apart from the field theory application. The crossing properties of conformal blocks forms a beautiful mathematical structure, so called the modular tensor category. This gives us a deep connection between 2D CFT and 3D TQFT. In fact, it can be shown explicitly from the Chern-Simons / WZW correspondence, which is a prototype of holographic duality.

More amazingly, the conformal blocks can also be view as a holomorphic section of a vector bundle over the moduli space of Riemann surfaces. This gives a deep connection between 2D CFT and complex geometry. And in fact this is deeply related to mathematical objects such as modular tensor category and modular functor.

1.2 Fundamental CFT

1.3 Outline

2 BPZ Conformal Blocks

2.1 Four Point Conformal Blocks

2.2 Torus Conformal Blocks

3 General Characterization of Conformal Blocks

3.1 Mathematical Definition

Its hard to generalize the definition of conformal blocks to general Riemann surfaces in a BPZ style. We need to find a more abstract characterization compatible with the 2 cases in the above section.

Mathematical definition may not be calculational enough, we need a more physical characterization of conformal blocks. There are generally 2 ways of viewing this I know, one is the chiral vertex operator construction, and the other is the sewing construction.

3.2 Chiral Vertex Operator Construction

3.3 Sewing Construction

Bibliography